1. The Core Design Philosophy

The Digital Library of Indic Intellectual History (DLIIH) interface must move away from the traditional "digital book" format. The UI will utilize a **Layered Disclosure Model**. Users are initially presented with an uncluttered, beautiful reading experience (Samhitapatha and Fluid Translation). Advanced philological data (Padapatha, grammatical tags, commentary) remains hidden until the user explicitly interacts with the text. This prevents cognitive overload while providing immense depth on demand.

2. MVP Interface Layout: The "Rigveda Viewer"

The localhost MVP will be a single-page application (SPA) divided into three dynamic interactive zones:

A. The Primary Reading Canvas (Center Pane)

- **Typography:** Large, highly legible Unicode Devanagari font that perfectly renders Vedic *svaras* (e.g., Noto Sans Devanagari, Siddhanta).
- **Structure:** Each mantra is presented as a distinct visual block.
- **Default View:** The accented *Samhitapatha* sits directly above the DLIIH fluid English translation.
- **Interaction:** Every single Sanskrit word in the Samhitapatha is a clickable UI element (rendered as an inline button or highlighted span on hover).

B. The Philological Deep-Dive Inspector (Right Sidebar / Sliding Drawer)

- When a user clicks a word in the Primary Canvas (e.g., `अ॒ग्निमी॑ळे`), this pane dynamically populates with the data mapped from our JSON schema:
  - **The Padapatha Break:** Shows the un-sandhied words (e.g., `अ॒ग्निम्` | `ई॒ळे`).
  - **Grammar Matrix:** Clean pill-shaped UI tags showing the morphological breakdown (`noun`, `masc`, `acc`, `sg`).
  - **Literal Meaning:** The raw semantic translation for just that word.
- *Broader Picture Goal:* In future phases, this pane will also display AI-generated cross-canonical parallels (e.g., linking a Vedic root to its Pali equivalent in early Śramaṇa texts).

C. The Traditional Apparatus (Bottom Accordion / Tab)

- Beneath the primary canvas, a toggleable section labeled "Classical Commentary."
- Expanding this reveals Sāyaṇa's *Bhāṣya* linked directly to the active mantra.
- *Broader Picture Goal:* This section will eventually scale to hold multiple commentaries (Madhva, Uvata, etc.) side-by-side.

3. Mobile-First "Pre-Seed" Considerations

Because our initial funding and user acquisition will be driven by Twitter/X and YouTube, a massive percentage of early traffic will be on mobile.

- The split-pane design must gracefully collapse into a bottom-sheet (like a modern map app) on mobile screens, where clicking a Sanskrit word slides the grammatical data up from the bottom of the phone screen.

4. Technical Stack (Localhost to Production)

To execute this HCI vision efficiently, the frontend will be built on a modern JavaScript framework.

- **Framework:** Next.js (React) or SvelteKit. These allow for rapid UI development and handle JSON data routing exceptionally well.
- **Styling:** Tailwind CSS. This utility-first framework allows us to quickly prototype beautiful, responsive UI components (like the grammar pills and sidebars) without writing thousands of lines of custom CSS.
- **Hosting Context:** Initially running via `localhost:3000` for development, later deployable to Vercel or AWS with zero configuration changes.